

HUBERT XU

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EDUCATION

University of Toronto
Bachelor's of Computer Science
Deans List Recipient (2023)

September 2022 - June 2027

SKILLS

Languages: Python, TypeScript, Java, C, C++, SQL
Frameworks: FastAPI, ExpressJS, Spring Frameworks, Next.js
Maching Learning: CUDA, PyTorch, NumPy, OpenCV, OpenClaw, MCP
Cloud: Google Cloud Platform, Microsoft Azure, AWS, Cloudflare,

EXPERIENCE

Amazon
Software Developer

Vancouver, BC
May 2026 – July 2026

MedMe Health (YC W21)
Software Engineer

Toronto, ON
May 2025 – December 2025

- Architected and deployed a production robotic process automation (RPA) system enabling bidirectional data synchronization across third-party healthcare platforms to support U.S. enterprise onboarding for Fortune 500 customers, directly enabling a **7 figure annual contract pipeline**.
- Built a resilient automation pipeline combining **vision-language models** for data extraction, UI automation, and template-based visual navigation, increasing successful happy-path workflows by **40%** and decreasing costs by **70%** while gracefully handling dynamic layouts and edge-case failures.
- Implemented end-to-end observability and performance metrics across automation services, **reducing SEV response time by 300%** and enabling proactive detection of bottlenecks impacting enterprise reliability.

UofTHacks 
President

Toronto, ON
May 2024 – Present

- Scaled the application platform to support **3,500+ applicants** and **800+ concurrent participants**.
- Managed and mentored a team of **40+ executives**, delegating tasks and fostering a collaborative culture.
- Designed, deployed, and operated the platform using **Next.js**, **FastAPI**, and **Cloudflare**, implementing caching, request routing, and edge optimizations to meet real-world latency and reliability constraints.
- Improved end-to-end platform performance and Core Web Vitals (LCP, TTI), contributing to a **184% increase in applicants** year-over-year.

PROJECTS

npm (Open Source Contributor) 

JavaScript

- Identified and fixed a critical vulnerability resolution bug in **metavuln-calculator** used by **npm audit**, restoring detection for prerelease package versions and improving security coverage across the npm ecosystem.

Kasli-OS 

C++, Linux, systemd

- Built a local first AI Linux management layer with a **C++ daemon**, **Unix socket IPC**, and TUI/CLI interface for structured OS inspection.
- Implemented typed read only tools, policy enforcement, audit logging, evidence redaction, and optional Ollama/LM Studio integration to let LLMs explain system state without privileged shell access.

UserThreads

C, Linux

- Implemented a user-level threading library for Ubuntu Linux featuring **preemptive scheduling** and priority-based thread management.
- Designed thread lifecycle management and context switching without relying on **makecontext** or **swapcontext**, demonstrating deep understanding of low-level concurrency and OS abstractions.

CERTIFICATIONS

- Microsoft Certified: Azure Fundamentals (AZ-900)